BIO301 -Assignment 2 Solution

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MSc Zoology

SEMESTER 1

COURTESY OF VIRTUAL UNIVERSITY OF PAKISTAN®

Q. 1 What are the limitations of Mendelian genetics? Explain (5 marks)

Limitations of Mendelian Genetics

The main limitation of Mendel's Principle is due to the following facts

1. It is not dominant in all cases of opposite characters. :

2. The type of combinatorial inheritance can appear in many crosses.

3. Sometimes two dominant combinations form a co-dominant, S. For example, when inheriting blood Group (MN Group)

4. Non-allelic interactions, such as the presence of epistasis, inhibitory factors, complementary genes and additional factors.

5. The relationship between different genes.

6. During gamete formation (that is, during meiosis) chromosomes do not diverge (lack of separation of homologous chromosomes).

7. Multiple factors (genes), which may or may not be alleles, are used for characteristics (such as a person's skin color, height, and weight) and their inheritance.

8. Gene pleiotropy, that is, genes that affect multiple traits.

9. Alleles, that is, there are a large number of alternative genes in a specific position on a chromosome.

10.Different degrees of penetrance (complete or incomplete) and the severity of specific genes at different times.

11. The existence of lethal genes, even the alleles of the genes that cause the organism to die

. 12. The influence of different environments on the genetic effects of a certain phenotype.

Q. 2 The gene for plant height in sunflowers has two co-dominant alleles, T1 for tall plants and T2 for short plants. If a tall plant is crossed with a short plant, Draw table and also write Genotype and Phenotype. (5 marks)

		Tall parent	
		T_1	T_1
Short parent	T_2	T_1T_2	T_1T_2
	T_2	T_1T_2	T_1T_2

Genotype:

T_1T_2(All Heterozygous Co-dominant)

Phenotype:

Medium Height (100%)

Remember in Prayers

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